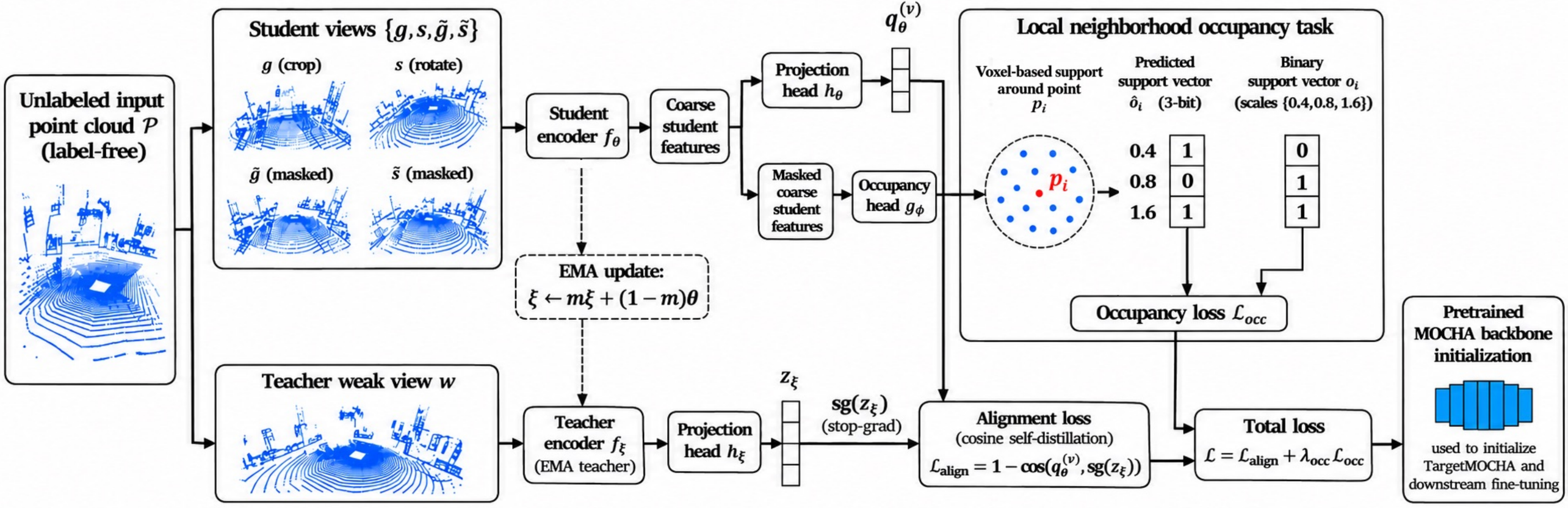

Unlabeled input point cloud $\mathcal{P}$ (label-free)
Student views $\{g, s, \tilde{g}, \tilde{s}\}$
g (crop)
s (rotate)
$\tilde{g}$ (masked)
$\tilde{s}$ (masked)
Teacher weak view w
Student encoder f_θ
Coarse student features
Projection head h_θ
$q_\theta^{(v)}$
Masked coarse student features
Occupancy head g_ϕ
EMA update:
$\xi \leftarrow m\xi + (1-m)\theta$
Teacher encoder f_ξ
(EMA teacher)
Projection head h_ξ
z_ξ
$\mathrm{sg}(z_\xi)$
(stop-grad)
Local neighborhood occupancy task
Voxel-based support around point p_i
Predicted support vector $\hat{o}_i$ (3-bit)
Binary support vector o_i (scales {0.4, 0.8, 1.6})
p_i
0.4
0.8
1.6
1
0
1
0
1
1
Occupancy loss $\mathcal{L}_{occ}$
Alignment loss
(cosine self-distillation)
$\mathcal{L}_{\text{align}} = 1 - \cos(q_\theta^{(v)}, \mathrm{sg}(z_\xi))$
Total loss
$\mathcal{L} = \mathcal{L}_{\text{align}} + \lambda_{occ}\mathcal{L}_{occ}$
Pretrained MOCHA backbone initialization
used to initialize TargetMOCHA and downstream fine-tuning